Adelmo Brunelli

Robotics Software Engineer | Autonomy, Real-Time Systems & Sim-to-Real

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Profile

Robotics-oriented software engineer with an MSc in Artificial Intelligence and Data Engineering. Research experience in **neuro-symbolic reinforcement learning** for legged robots (**Unitree Go2 Pro**, **Nvidia Isaac Sim**) with **sim-to-real** transfer on physical hardware, plus two years of industrial work on **hard real-time**, deterministic and fault-tolerant systems for safety-critical applications. Comfortable from low-level **C/C++** and real-time communication up to learning-based control in **Python**.

Experiences

Artificial Intelligence Research Assistant — Robotics London, UK
Imperial College London (2^{nd} World University 2026 - QS Ranking) *Mar. 2024 – Mar. 2025*

- Designed and implemented a novel algorithm for **Neuro-Symbolic Reinforcement Learning** to train a **Unitree Go2 Pro quadruped** on complex **Non-Markovian**, temporally extended tasks.
- Built the full pipeline in the **Nvidia Isaac Simulator**: environment and reward specification, policy training, evaluation, and **sim-to-real transfer** validated on the physical robot.
- Joined the *SPIKE Group*, led by Alessandra Russo, with **quantitative validation** of policy performance and of the simulation-to-hardware gap.
- Project part of an international collaboration with **University of Cardiff**, **University of Brescia**, and **UCLA**, funded by **DEVCOM Army Research Laboratory**.

Advanced R&D Software Engineer La Spezia, IT
Leonardo Global Solutions - Aerospace & Defense *May 2026 – Present*

- Research on reliable Ethernet-based communications, ensuring **deterministic, low-latency and fault-tolerant** data exchange under **real-time constraints** — the same requirements that govern distributed robot control loops and sensor buses.
- Design, integration and **on-target validation** of advanced terrestrial and naval defense systems, working on complex **mission-critical** architectures that meet stringent **reliability, safety** and operational requirements.

Embedded Software Engineer La Spezia, IT
Capgemini Engineering *Mar 2025 – Apr. 2026*

- Development and analysis of **hard real-time (HRT)** systems and low-latency architectures for deterministic communication in mission-critical environments.
- In-depth work on **IEEE** standards, real-time networking protocols and **low-level programming**, with characterization of **latency variance and jitter** in high-performance systems.

Software Engineer Remote
Amazon *Sep. 2020 – Jan. 2021*

- Developed an **Amazon Alexa Skill** integrated with **AWS** (remote invocation via **HTTPS** and the **Alexa Skills Kit**), with backend logic in **Node.js** and **SQL/MySQL**.

Education

University of Pisa Pisa, IT
Master of Science - MsC, Artificial Intelligence and Data Engineering *Feb. 2025*

- **Thesis**: *Reward Machines for Real World Navigation* — temporally extended task specification for **robot navigation**. *Supervisors: A. Russo, M. Cimino*
- Specialized in **AI Robotics**, Deep Learning, Statistical Modeling and Data Engineering.

University of Pisa Pisa, IT
Bachelor of Applied Science - BASc, Computing Engineering *Jan. 2022*

- Focus on Algorithms, Software Engineering and **Embedded Systems**. **Thesis**: Preliminary analysis of new Twitter account behavior. *Supervisor: M. Avvenuti*

Technical Skills

Robotics & Simulation: *Nvidia Isaac Simulator, legged/quadruped platforms (Unitree Go2 Pro), sim-to-real transfer, Reinforcement Learning, Reward Machines, neuro-symbolic methods, Matlab.*
Real-Time & Embedded: *hard real-time (HRT) systems, deterministic Ethernet, IEEE real-time networking protocols, low-level and low-latency programming, safety- and mission-critical validation.*
Programming: *C/C++, Python, Rust, Java/Scala, Bash, SQL (Postgres), Julia, R.*
ML & Data: *TensorFlow, Keras, Scikit-Learn, Numpy, Pandas, HuggingFace, MLflow, Apache Spark, AWS SageMaker, GCP Vertex AI.*
Developer Tools: *Git, Docker, Kubernetes, TensorBoard, VS Code, Visual Studio, IntelliJ, Google Colab.*
Evaluation & Analysis: *Hypothesis Testing, Confidence Intervals, ANOVA, Regression Analysis, Bayesian Inference, Time Series Forecasting.*
Spoken Languages: *English — IELTS UKVI Band 7.0, C1* | *Italian — Native* | *German — Basic.*